SECTION A: MULTIPLE CHOICE QUESTIONS - 20 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. START WITH SECTION A AND HAND THE OPTIC READER FORM IN AFTER 75 MINUTES.
2. First circle your answers on this paper and only mark the answers on the form when you are certain of your choice. You may not erase on the optic reader form.
3. Do all scribbling on the facing page. It will not be marked.
4. Use a soft pencil to write on the optic reader form.
5. Use **SIDE 2** of the optic reader form.
   Fill in your personal information in pencil on side 2.
   Fill in your student number from top to bottom and then code it.
6. **You may start with section B as soon as you have marked your answers on the optic reader form.**

AFDELING A : MEERVOUDIGE KEUSEVRAE - 20 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. BEGIN MET AFDELING A EN HANDIG DIE MERKLEESVORM IN NA 75 MINUTE.
2. Omkring jou antwoorde eers in hierdie vraestel en merk die antwoorde op die merkleesvorm vorm eers as jy seker is van jou keuse. Jy mag nie uitvee op die merkleesvorm nie.
3. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
4. Gebruik 'n sagte potlood om op die merkleesvorm te skryf.
5. Gebruik **KANT 2** van die merkleesvorm.
   Vul u persoonlike inligting in potlood in op kant 2.
   Vul u studentenommer van bo na onder in en kodeer dit.
6. **U kan met afdeling B begin sodra u die antwoorde op die merkleesvorm ingevul het.**

Question 1 / Vraag 1

If $\dfrac{(e^{3x})^2 e^{2x}}{e^{5x}} = e^a$, then $a =$ / As $\dfrac{(e^{3x})^2 e^{2x}}{e^{5x}} = e^a$, dan is $a =$

(a)	$3x$	(b)	$9x^2 - 3x$	(c)	$\frac{18}{5}x^2$
(d)	$-3x$	(e)	$9x^2$	(f)	None of these / Geen van hierdie

Question 2 / Vraag 2

The solution of $|3x - 1| \geq 5$ is

Die oplossing van $|3x - 1| \geq 5$ is

(a)	$x \geq -\frac{4}{3}$ or/of $x \leq 2$	(b)	$x \leq -\frac{4}{3}$ or/of $x \geq 2$	(c)	$x \geq 2$ or/of $x \leq \frac{4}{3}$
(d)	$x \leq -2$ and/en $x \geq \frac{4}{3}$	(e)	$x \geq -\frac{4}{3}$ and/en $x \leq 2$	(f)	None of these / Geen van hierdie

Question 3 / Vraag 3

If $-x^2 + 4x - 3 > 0$, then

Indien $-x^2 + 4x - 3 > 0$, dan is

(a)	$x \in (0,3)$	(b)	$x \in [1,3)$	(c)	$x = 2$
(d)	$x \in (1,3)$	(e)	$x \in (-\infty,1) \cup (3,\infty)$	(f)	None of these / Geen van hierdie

Question 4 / Vraag 4

If $\frac{x-2}{|x-3|} \geq 0$, then

Indien $\frac{x-2}{|x-3|} \geq 0$, dan is

(a)	$x \in [2,\infty)$	(b)	$x \in (2,3) \cup (3,\infty)$	(c)	$x \in [2,3) \cup (3,\infty)$
(d)	$x \in (-\infty,3)$	(e)	$x \in (-\infty,2) \cup (2,3)$	(f)	None of these / Geen van hierdie

Question 5 / Vraag 5

If $f(x) = 2x^2 + 1$ and $g(x) = x - 2$, then $(f \circ g)(2) =$

Indien $f(x) = 2x^2 + 1$ en $g(x) = x - 2$, dan is $(f \circ g)(2) =$

(a)	-3	(b)	9	(c)	4
(d)	1	(e)	33	(f)	None of these / Geen van hierdie

Question 6 / Vraag 6

If / Indien $f(x) = \begin{cases} |x| & \text{if/as} \quad x < 2 \\ x-3 & \text{if/as} \quad x \geq 2 \end{cases}$, then / dan is

$f(-3) =$

(a)	-3	(b)	3	(c)	0
(d)	2	(e)	-1	(f)	None of these / Geen van hierdie

Question 7 / Vraag 7

$\tan(-\frac{7\pi}{6}) =$

(a)	$-\sqrt{3}$	(b)	-1	(c)	$\frac{1}{\sqrt{3}}$
(d)	$\sqrt{3}$	(e)	$-\frac{1}{\sqrt{3}}$	(f)	None of these / Geen van hierdie

Question 8 / Vraag 8

$\text{cosec}(\frac{47\pi}{2}) =$

(a)	1	(b)	-1	(c)	0
(d)	$\frac{\sqrt{3}}{2}$	(e)	$-\frac{2}{\sqrt{2}}$	(f)	None of these / Geen van hierdie

Question 9 / Vraag 9

For which value(s) of x is $\tan(\tan^{-1}x) = x$?

Vir watter waarde(s) van x is $\tan(\tan^{-1}x) = x$?

(Notation / Notasie: $\tan^{-1} = \arctan = bg\tan$)

(a)	$0 \leq x \leq \pi$	(b)	$-1 \leq x \leq 1$	(c)	$-\pi \leq x \leq \pi$
(d)	$-\frac{\pi}{2} < x < \frac{\pi}{2}$	(e)	$x \in \mathbb{R}$	(f)	None of these / Geen van hierdie

Question 10 / Vraag 10

If / As $\ln x + \ln(x+2) = 0$, then / dan is

(a)	$x = 0$ or/of $x = -2$	(b)	$x = -1$	(c)	$x = -1 \pm \sqrt{2}$
(d)	$x = -1 + \sqrt{2}$	(e)	$x = 1$	(f)	None of these / Geen van hierdie

Question 11 / Vraag 11

The equation of the graph obtained by reflecting the graph of $y = |x-1|$ about the y-axis, stretching it vertically by a factor of 2 and translating it down by 3 is given by:

Die vergelyking van die grafiek wat verkry word deur die grafiek van $y = |x-1|$ om die y-as te reflekteer, vertikaal te rek met 'n faktor van 2 en dan af te skuif met 3 eenhede word gegee deur:

(a)	$y = 2\|x+1\| - 3$	(b)	$y = \|x+1\| - 3$	(c)	$y = -2\|x+1\| + 3$
(d)	$y = \|x-1\| - 3$	(e)	$y = 2\|x+1\| + 3$	(f)	None of these / Geen van hierdie

Question 12 / Vraag 12

If $xe^{-x} + 2e^{-x} = 0$, then

Indien $xe^{-x} + 2e^{-x} = 0$, dan is

(a)	$x = -2$	(b)	$x = -2$ or / of $x = 0$	(c)	$x = 0$ or / of $x = 1$
(d)	$x = 0$	(e)	$x = -2$ or / of $x = 1$	(f)	None of these / Geen van hierdie

Question 13 / Vraag 13

If $\ln a = r$, $\ln b = s$ and $\ln c = t$, then $\ln\left(b\sqrt{\frac{a}{c^2}}\right) =$

Indien $\ln a = r$, $\ln b = s$ en $\ln c = t$, dan is $\ln\left(b\sqrt{\frac{a}{c^2}}\right) =$

(a)	$2r + 2s - \frac{1}{2}t$	(b)	$r + s + \frac{1}{2}t$	(c)	$\frac{1}{2}(r+s) - t$
(d)	$\frac{1}{2}r + s - t$	(e)	$r + \frac{1}{2}s + t$	(f)	None of these / Geen van hierdie

Question 14 / Vraag 14

The inverse function to $f(x) = \frac{x+1}{x+2}$ is $f^{-1}(x) =$

Die inverse funksie van $f(x) = \frac{x+1}{x+2}$ is $f^{-1}(x) =$

(a)	$\frac{x-1}{x-2}$	(b)	$\frac{2x-1}{x-2}$	(c)	$\frac{x-1}{2x-1}$
(d)	$\frac{1-2x}{x-1}$	(e)	$\frac{1-2x}{x-2}$	(f)	None of these / Geen van hierdie

Question 15 / Vraag 15

The domain of the function $f(x) = \frac{1}{1 + \ln(x+1)}$ is

Die definisieversameling van die funksie $f(x) = \frac{1}{1 + \ln(x+1)}$ is

(a)	$\left(-1 + \frac{1}{e}, \infty\right)$	(b)	$(-1, \infty)$	(c)	$\left(-1, -1 + \frac{1}{e}\right) \cup \left(-1 + \frac{1}{e}, \infty\right)$
(d)	$\left(-1, -1 + \frac{1}{e}\right)$	(e)	$x \in \mathbb{R},\ x \neq -1 + \frac{1}{e}$	(f)	None of these / Geen van hierdie

Question 16 / Vraag 16

For which value(s) of x is $y = \sin^{-1}(2x - 1)$ defined?

Vir watter waarde(s) van x is $y = \sin^{-1}(2x-1)$ gedefinieer?

(Notation / Notasie: $\sin^{-1} = \arcsin = bg\sin$)

(a)	$0 \leq x \leq 1$	(b)	$-1 \leq x \leq 1$	(c)	$-\frac{\pi}{4} + \frac{1}{2} \leq x \leq \frac{\pi}{4} - \frac{1}{2}$
(d)	$-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$	(e)	$x \in \mathbb{R}$	(f)	None of these / Geen van hierdie

Question 17 / Vraag 17

$\lim\limits_{x \to 5} \operatorname{cosec} \frac{\pi x}{4} =$

(a)	1	(b)	-1	(c)	$-\sqrt{2}$
(d)	$-\frac{1}{\sqrt{2}}$	(e)	$\sqrt{2}$	(f)	None of these / Geen van hierdie

Question 18 / Vraag 18

$$\lim_{x\to a} \frac{x^2 - 2ax + a^2}{x^2 - a^2} =$$

(a)	$\frac{1}{2a}$	(b)	$-\frac{1}{2a}$	(c)	$-\frac{1}{2a^2}$
(d)	$\frac{1}{2a^2}$	(e)	0	(f)	None of these / Geen van hierdie

Question 19 / Vraag 19

$$\lim_{x\to 0} \frac{\sin 2x \tan x}{x^2} =$$

(a)	0	(b)	1	(c)	2
(d)	π	(e)	-1	(f)	None of these / Geen van hierdie

Question 20 / Vraag 20

Given / Gegee: $\overline{a} = \langle 1,2,-3\rangle$ and/en $\overline{b} = \langle 4,0,7\rangle$.
Expressed as standard basis vectors, $2\overline{a} + 3\overline{b} =$

Uitgedruk as standaard basis vektore, is $2\overline{a} + 3\overline{b} =$

(a)	$8\overline{i} - \overline{j} + 4\overline{k}$	(b)	$8\overline{i} + \overline{j} + 4\overline{k}$	(c)	$8\overline{i} - \overline{j} + 15\overline{k}$
(d)	$4\overline{i} + 14\overline{j} + 15\overline{k}$	(e)	$14\overline{i} + 4\overline{j} + 15\overline{k}$	(f)	None of these / Geen van hierdie

20

SECTION B: 25 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. Answer the following questions on this paper **in ink**.
2. Work done in pencil or in red ink will not be marked.
3. Do not use correction fluid ("Tipp-Ex").
4. Do all scribbling on the facing page. It will not be marked.
   If you need more space for an answer, use the facing page and indicate it clearly **by framing it**.
5. Show all steps and calculations and explain your answers.

AFDELING B: 25 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. Beantwoord die volgende vrae op hierdie vraestel **in ink**.
2. Werk wat in potlood of rooi ink beantwoord is, word nie nagesien nie.
3. Korrigeer-vloeistof ("Tipp-Ex") mag nie gebruik word nie.
4. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
   As jy nog ruimte vir 'n antwoord nodig het, gebruik die teenblad en dui dit duidelik aan **deur 'n raam daarom te trek**.
5. Toon alle stappe en berekeninge en verduidelik jou antwoorde.

Question 1 / Vraag 1

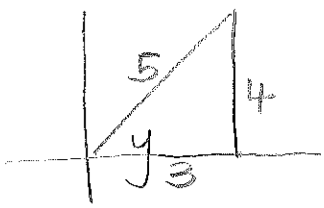
Find the value of / Vind die waarde van

$$\sin\left[2\cos^{-1}\left(\tfrac{3}{5}\right)\right]$$

(Notation / Notasie: $\cos^{-1}$ = arccos = $bg\cos$)

Let $y = \cos^{-1}\left(\frac{3}{5}\right)$

$\Rightarrow \cos y = \frac{3}{5}$ and $y \in [0, \pi]$



So $\sin 2y$

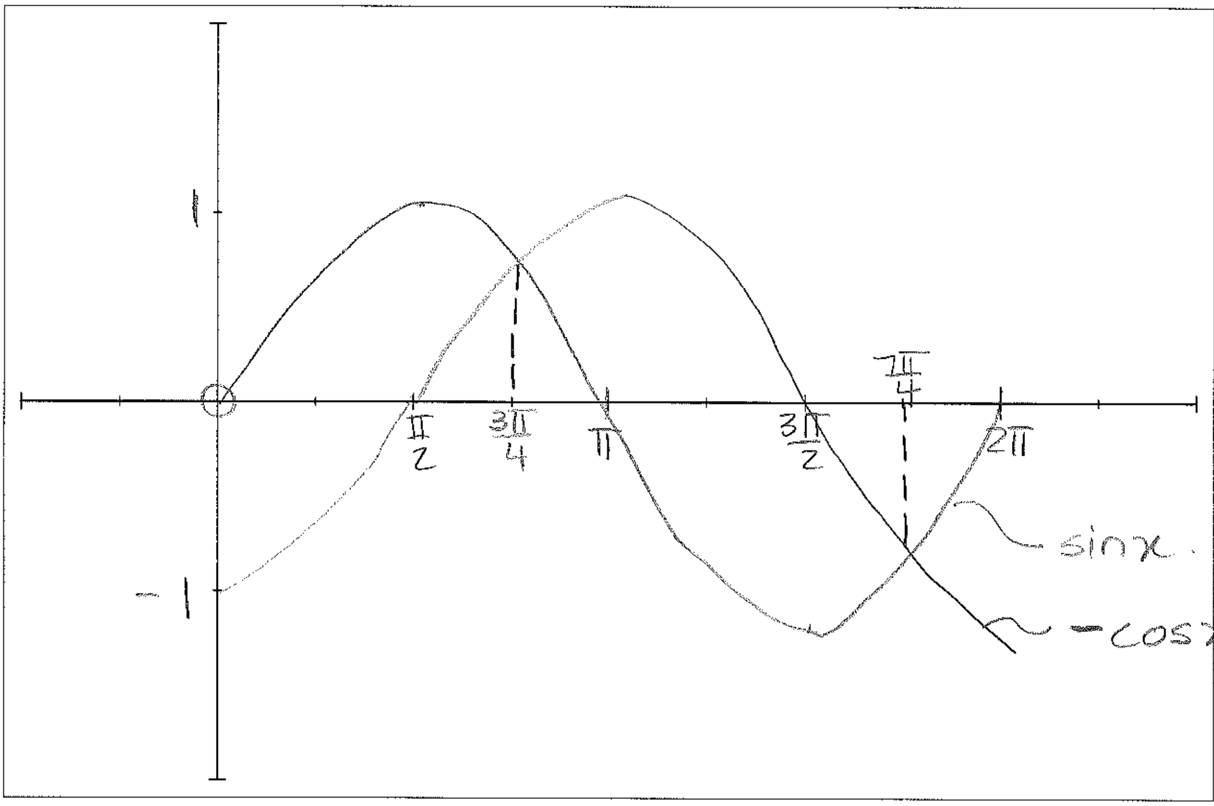
$= 2\sin y \cos y$

$= 2\left(\frac{4}{5}\right)\left(\frac{3}{5}\right) = \frac{24}{25}$

3

Question 2 / Vraag 2

i. Sketch the graphs (in ink) of $y = \sin x$ and $y = -\cos x$ on the same set of axis if $x \in [0, 2\pi]$. (Remember to indicate all the relevent information on the graph.)
Skets die grafieke (in ink) van $y = \sin x$ en $y = -\cos x$ op dieselfde assestelsel indien $x \in [0, 2\pi]$. (Onthou om alle belangrike inligting op die grafiek aan te dui.)



ii. Use the graph to solve for x if $\sin x \geq -\cos x$.

Gebruik die grafiek en los vir x op as $\sin x \geq -\cos x$.

Consider $\sin x = -\cos x \Rightarrow \tan x = -1$
(RA : $\frac{\pi}{4}$)

So $x = \frac{3\pi}{4}$ or $\frac{7\pi}{4}$ (2^{nd} / 4^{th} quadrant)

$\sin x \geq -\cos x$ if

$x \in \left[0, \frac{3\pi}{4}\right] \cup \left[\frac{7\pi}{4}, 2\pi\right]$

4

Question 3 / Vraag 3

i. Use the definition of the absolute value to write $f(x) = |x + 5|$ without the absolute value symbol.

Gebruik die definisie van die absolute waarde om $f(x) = |x + 5|$ sonder die absolute waarde simbool te skryf.

$$f(x) = |x+5| = \begin{cases} x+5 & \text{if} \quad x \geq -5 \\ -(x+5) & \text{if} \quad x < -5 \end{cases}$$

1

ii. Determine / Bepaal $\lim\limits_{x\to-5} \dfrac{|x+5|}{x+5}$.

$$\lim_{x \to -5^-} \frac{-(x+5)}{x+5}$$

$$= -1$$

$$\lim_{x \to -5^+} \frac{x+5}{x+5}$$

$$= 1$$

$$\Rightarrow \lim_{x \to -5} \frac{|x+5|}{x+5} \quad \text{DNE.}$$

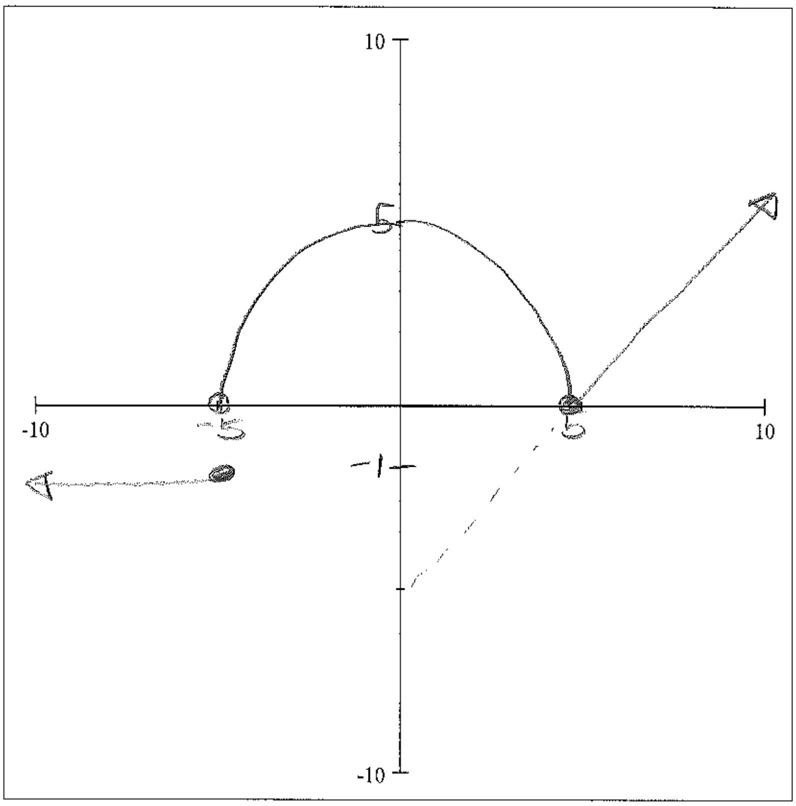
2

Question 4 / Vraag 4

Sketch (in ink) the graph of the given function. (Remember to indicate all the relevent information on the graph.)

Skets (in ink) die grafiek van die gegewe funksie. (Onthou om alle belangrike inligting op die grafiek aan te dui.)

$$f(x) = \begin{cases} -1 & , \quad x \leq -5 \\ \sqrt{25 - x^2} & , \quad -5 < x < 5 \\ x - 5 & , \quad x \geq 5 \end{cases}$$



2

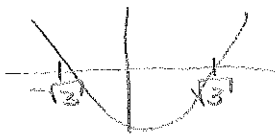
Question 5 / Vraag 5

Solve the inequality / Los die ongelykheid op

$$\ln(x^2 - 3) \leq 0$$

$\ln(x^2-3) \leq 0$ and $x^2 - 3 > 0$

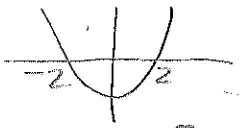
$\Rightarrow (x-\sqrt{3})(x+\sqrt{3}) > 0$



$x \in (-\infty, -\sqrt{3}) \cup (\sqrt{3}, \infty)$

$\Rightarrow x^2 - 3 \leq 1$

$\Rightarrow x^2 - 4 \leq 0$



$\Rightarrow x \in [-2, 2]$

$\Rightarrow x \in [-2, -\sqrt{3}) \cup (\sqrt{3}, 2]$

4

Question 6 / Vraag 6

Find the vertical asymptote of $f(x) = \frac{x+2}{x+3}$, if it exist. Show all your calculations.

Vind die vertikale asimptoot van $f(x) = \frac{x+2}{x+3}$, indien dit bestaan. Toon aan al jou berekeninge.

$D_f = \mathbb{R} \setminus \{-3\}$. So $x = -3$ is a possible V.A.

$$\lim_{x \to -3^-} \frac{x+2}{x+3} = +\infty$$

OR

$$\lim_{x \to -3^+} \frac{x+2}{x+3} = -\infty$$

Therefore $x = -3$ is a V. A

3

Question 7 / Vraag 7

Given points $A(2,3,4)$ and $B(10,2,8)$. Find a vector parallel to $\overline{AB}$ with length 18.

Gegee punte $A(2,3,4)$ en $B(10,2,8)$. Vind 'n vektor parallel aan $\overline{AB}$ met lengte 18.

$\overline{AB} = \langle 8, -1, 4 \rangle$

$|\overline{AB}| = \sqrt{64+1+16} = \sqrt{81} = 9$

$\Rightarrow \frac{\overline{AB}}{|\overline{AB}|} = \frac{\langle 8, -1, 4 \rangle}{9}$

Vector $\parallel$ to $\overline{AB}$ with length 18

is $18 \times \frac{\langle 8, -1, 4 \rangle}{9}$ 3

$= \langle 16, -2, 8 \rangle$

Question 8 / Vraag 8

i. Write down the definition of an odd function. / Gee die definisie van 'n onewe funksie.

f is odd if

$f(-x) = -f(x)$ for all $x \in D_f$

1

ii. Show that / Toon aan dat

$$f(x) = \ln(x + \sqrt{x^2 + 1})$$

is an odd function. / 'n onewe funksie is.

$f(-x) = \ln(-x + \sqrt{x^2+1})$ and.

$$-f(x) = -\ln(x + \sqrt{x^2+1})$$

$$= \ln\left[\frac{1}{x + \sqrt{x^2+1}} \times \frac{x - \sqrt{x^2+1}}{x - \sqrt{x^2+1}}\right]$$

$$= \ln\left[\frac{x - \sqrt{x^2+1}}{x^2 - (x^2+1)}\right]$$

3

$$= \ln\left[\frac{x - \sqrt{x^2+1}}{-1}\right]$$

$$= \ln\left[-x + \sqrt{x^2+1}\right] = f(-x)$$

$\Rightarrow f$ is odd.